MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, SCHOTTKY TTL,

MAGNITUDE COMPARATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Schottky TTL, magnitude comparator microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-bit comparator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
X	C-2A (20-terminal, .350" x .350"), square chip carrier package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	605 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}): Cases E, F, X, and 2	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc 3/
Normalized fanout (each output)	10 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth table. The truth table shall be as specified on figure 2.

3.2.3 Logic diagram. The logic diagram shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3/ 0.7 V at +125°C.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Limits		Unit
			Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V V_{IL} = 0.8 V, I_{OH} = -1 mA 1/	2.5		V
		T_C = +125°C, V_{IL} = 0.7 V			
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.8 V, I_{OL} = 20 mA 1/		0.5	V
		T_C = +125°C, V_{IL} = 0.7 V		0.45	
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C		-1.2	V
High-level input current (A < B, A > B inputs)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		50	µA
High-level input current (all other inputs)	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		150	µA
High-level input current (all inputs)	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		1	mA
Low-level input current (A < B, A > B inputs)	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	-1.0	-2.0	mA
Low-level input current (all other inputs)	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	-3.0	-6.0	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/ 2/	-40	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V 3/		115	mA
Collector cut-off current	I_{CEX}	V_{CC} = 5.5 V, V_{IL} = GND V_{OH} = 5.5 V		250	µA
Propagation delay from any A or B input to A < B or A > B outputs	t_{PLH1}	C_L = 50 pF ±10%, 3 gate levels R_L = 280Ω 4/ 5/	2	23	ns
Propagation delay from any A or B input to A = B output	t_{PLH2}	C_L = 50 pF ±10%, 3 gate levels R_L = 280Ω 4/ 5/	2	25	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
Propagation delay from A < B or A = B input to A > B output	t_{PLH3}	C_L = 50 pF ±10%, 2 gate levels R_L = 280Ω 4/ 5/	2	11.5	ns
Propagation delay from A = B input to A = B output	t_{PLH4}	C_L = 50 pF ±10%, 1 gate level R_L = 280Ω 4/ 5/	2	16	ns
Propagation delay from A > B or A = B input to A < B output	t_{PLH5}	C_L = 50 pF ±10%, 2 gate levels R_L = 280Ω 4/ 5/	2	11.5	ns
Propagation delay from any A or B input to A > B or A < B output	t_{PHL1}	C_L = 50 pF ±10%, 3 gate levels R_L = 280Ω 4/ 5/	2	24	ns
Propagation delay from any A or B input to A = B output	t_{PHL2}	C_L = 50 pF ±10%, 3 gate levels R_L = 280Ω 4/ 5/	2	24	ns
Propagation delay from A < B or A = B inputs to A > B output	t_{PHL3}	C_L = 50 pF ±10%, 2 gate levels R_L = 280Ω 4/ 5/	2	13	ns
Propagation delay from A = B input to A = B output	t_{PHL4}	C_L = 50 pF ±10%, 1 gate level R_L = 280Ω 4/ 5/	2	12	ns
Propagation delay from A > B or A = B input to A < B output	t_{PHL5}	C_L = 50 pF ±10%, 2 gate levels R_L = 280Ω 4/ 5/	2	13	ns

1/ All unspecified inputs grounded.

2/ Not more than one output should be shorted at a time.

3/ I_{CC} is measured with outputs open, A = B grounded, and all other inputs at 4.5 V.

4/ The longest gate level propagation delay paths are not tested (5 gate levels); however, they can be calculated: $t = tp_1$ maximum + tp_2 maximum - tp_4 minimum.

5/ Gate level references are made with respect to the circuit logic diagram.

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3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number II (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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Pin number		Pin name	
Cases		Device type 01	
2, X	E, F	2, X	E, F
1	1	NC	B3
2	2	B3	A < B In
3	3	A < B In	A = B In
4	4	A = B In	A > B In
5	5	A > B In	A > B Out
6	6	NC	A = B Out
7	7	A > B Out	A < B Out
8	8	A = B Out	GND
9	9	A < B Out	B0
10	10	GND	A0
11	11	NC	B1
12	12	B0	A1
13	13	A0	A2
14	14	B1	B2
15	15	A1	A3
16	16	NC	V_{CC}
17		A2	
18		B2	
19		A3	
20		V_{CC}	

FIGURE 1. Terminal connections.

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Device type 01

Comparing inputs				Cascading inputs			Outputs		
A3, B3	A2, B2	A1, B1	A0, B0	A > B	A < B	A = B	A > B	A < B	A = B
A3 > B3	X	X	X	X	X	X	H	L	L
A3 < B3	X	X	X	X	X	X	L	H	L
A3 = B3	A2 > B2	X	X	X	X	X	H	L	L
A3 = B3	A2 < B2	X	X	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 > B1	X	X	X	X	H	L	L
A3 = B3	A2 = B2	A1 < B1	X	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 > B0	X	X	X	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 < B0	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	H	L	L	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	H	L	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	L	H	L	L	H
A3 = B3	A2 = B2	A1 = B1	A0 = B0	X	X	H	L	L	H
A3 = B3	A2 = B2	A1 = B1	A0 = B0	H	H	L	L	L	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	L	L	H	H	L

H = High level
L = Low level
X = Irrelevant

FIGURE 2. Truth table.

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Device type 01

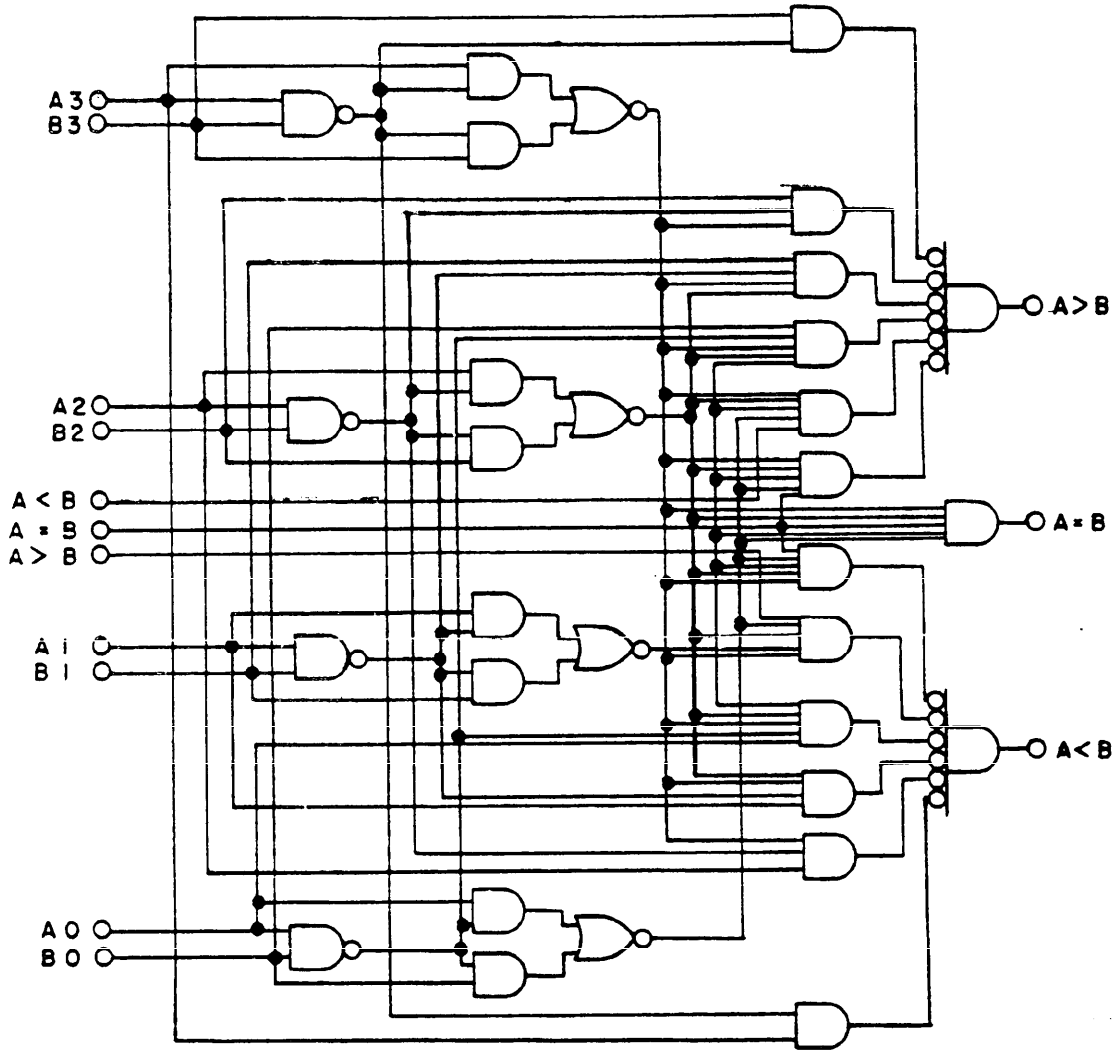
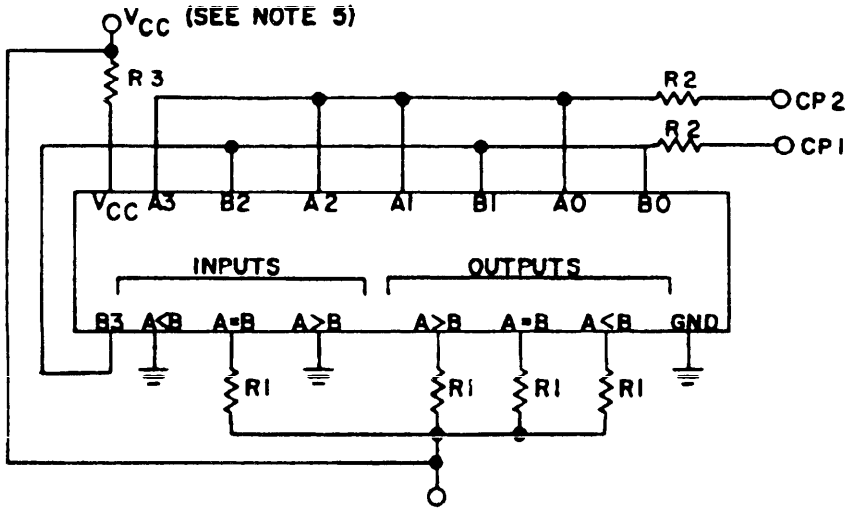


FIGURE 3. Logic diagram.

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Device type 01

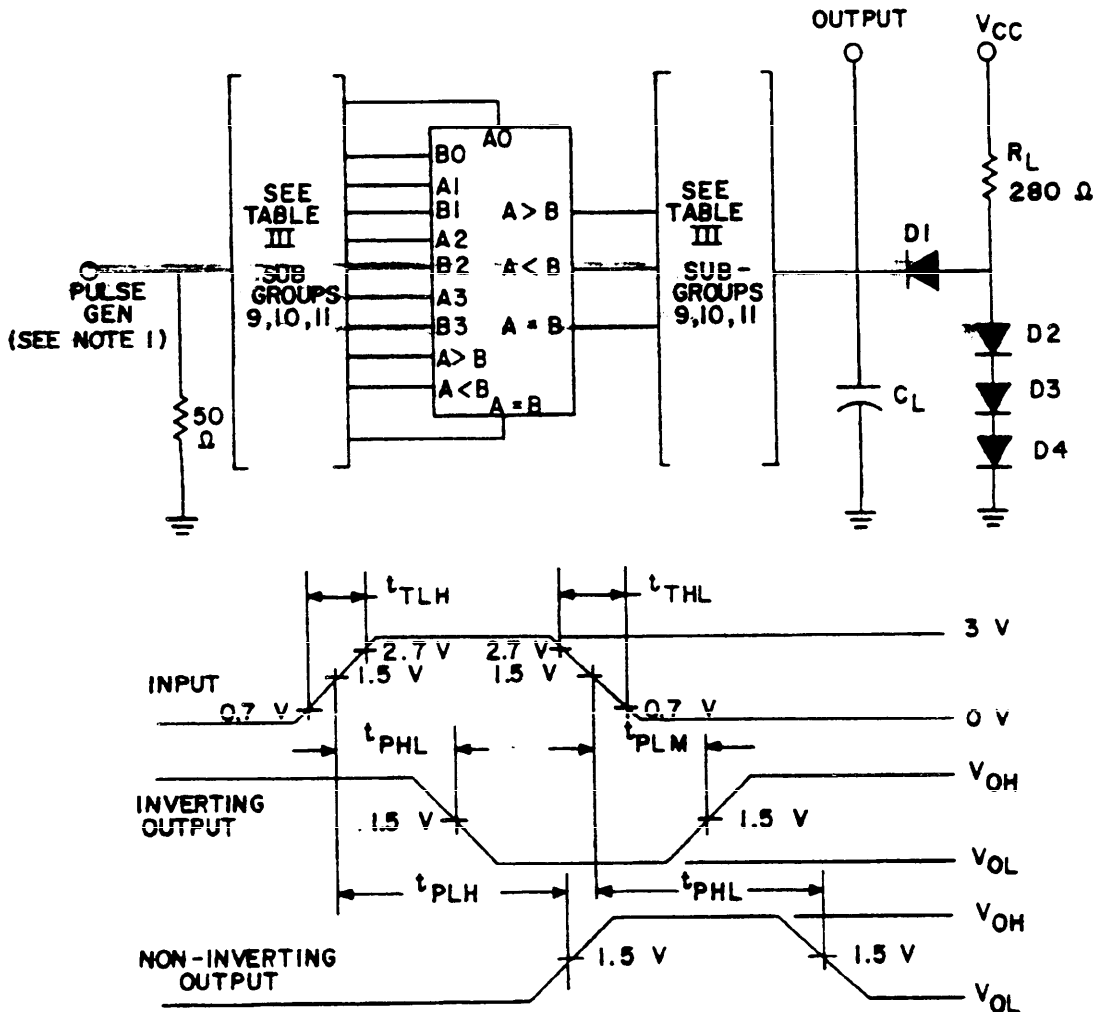


NOTES:
1. CP1 = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. CP2 = 50 kHz ±50 percent square wave synchronized with CP1. All other conditions are the same as CP1.
3. R1 = 220Ω ±5 percent.
4. R2 = 27Ω ±5 percent.
5. V_{CC} and R3 shall be chosen to insure 5.0 V minimum at the device terminal.

FIGURE 4. Burn-in and life test circuit.

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Device type 01



NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, t_{TLH} and t_{THL} ≤ 2.5 ns, t_p = 500 ns, PRR ≤ 1 MHz, and Z_{out} = 50Ω.
2. C_L = 50 pF ±10 percent and includes probe and jig capacitance.
3. All diodes are 1N3064, or equivalent.

FIGURE 5. Switching time test circuit.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases C and D / Cases 2 and 3	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ X and Z	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B_3	A < B IN	A = B IN	A > B IN	A > B OUT	A = B OUT	A < B OUT	GND	B_0	A_0	B_1	A_1	A_2	B_2	A_3	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	0.8 V	0.8 V		-1 mA			GND	0.8 V	GND	0.8 V	GND	GND	0.8 V	GND	4.5 V	A > B OUT	2.5		V dc
			2	0.8 V		2.0 V			-1 mA		"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	A = B OUT	"		"
			3	GND		0.8 V	0.8 V			-1 mA	"	GND	0.8 V	GND	0.8 V	0.8 V	GND	0.8 V	"	A < B OUT	"		"
	V_{OL}	3007	4	2.0 V	2.0 V	GND		20 mA			"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	A > B OUT		0.5	"
			5	2.0 V		0.8 V			20 mA		"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	A = B OUT		"	"
			6	0.8 V		2.0 V	GND			20 mA	"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	A < B OUT		"	"
	V_{IC}		7	-18 mA							"								"	B_3		-1.2	"
			8		-18 mA						"								"	A < B IN		"	"
			9			-18 mA					"								"	A = B IN		"	"
			10				-18 mA				"								"	A > B IN		"	"
			11								"	-18 mA							"	B_0		"	"
			12								"		-18 mA						"	A_0		"	"
			13								"			-18 mA					"	B_1		"	"
			14								"				-18 mA				"	A_1		"	"
			15								"					-18 mA			"	A_2		"	"
			16								"						-18 mA		"	B_2		"	"
			17								"							-18 mA	"	A_3		"	"
	I_{IH1}	3010	18	GND	2.7 V						"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	A < B IN		50	µA
			19	GND			2.7 V				"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A > B IN		50	"
	I_{IH2}		20	2.7 V							"							GND	"	B_3		150	"
			21	GND		2.7 V					"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A = B IN		"	"
			22	"							"	2.7 V	GND	"	"	"	GND	"	"	d_0		"	"
			23	"							"	GND	2.7 V	"	"	"	"	"	"	A_0		"	"
			24	"							"	"	5.5 V	2.7 V	"	"	"	"	"	B_1		"	"
			25	"							"	"	"	GND	2.7 V	"	"	"	"	A_1		"	"
			26	"							"	"	"	"	GND	2.7 V	"	"	"	A_2		"	"
			27	"							"	"	"	"	5.5 V	GND	2.7 V	"	"	B_2		"	"
			28	"							"							2.7 V	"	A_3		"	"
			29	2.7 V							"							5.5 V	"	B_3		"	"
			30	GND							"	2.7 V	5.5 V	GND	5.5 V	5.5 V	GND	"	"	B_0		"	"
			31	"							"	5.5 V	2.7 V	GND	5.5 V	"	"	"	"	A_0		"	"
			32	"							"	GND	5.5 V	2.7 V	GND	"	"	"	"	B_1		"	"
			33	"							"	"	"	5.5 V	2.7 V	"	"	"	"	A_1		"	"
			34	"							"	"	"	GND	5.5 V	2.7 V	5.5 V	"	"	A_2		"	"
			35	"							"	"	"	GND	5.5 V	5.5 V	2.7 V	"	"	B_2		"	"
			36	5.5 V							"							2.7 V	"	A_3		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, low <0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 1/ X and Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B_3	A < B IN	A = B IN	A > B IN	A > B OUT	A = B OUT	A < B OUT	GND	B_0	A_0	B_1	A_1	A_2	B_2	A_3	V_{CC}		Min	Max	
1 T_C = +25°C	I_{IH3}	3010	37	GND	5.5 V						GND	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	A < B IN		1	mA
			38	GND			5.5 V				"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A > B IN		"	"
			39	5.5 V							"							GND	"	B_3		"	"
			40	GND		5.5 V					"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A = B IN		"	"
			41	"							"	5.5 V	GND	"	"	"	GND	"	"	B_0		"	"
			42	"							"	GND	5.5 V	"	"	"	"	"	"	A_0		"	"
			43	"							"	"	"	"	"	"	"	"	"	B_1		"	"
			44	"							"	"	"	GND	"	"	"	"	"	A_1		"	"
			45	"							"	"	"	"	GND	"	"	"	"	A_2		"	"
			46	"							"	"	"	"	5.5 V	GND	5.5 V	"	"	B_2		"	"
			47	"							"							"	"	A_3		"	"
			48	5.5 V							"							"	"	B_3		"	"
			49	GND							"	5.5 V	5.5 V	GND	5.5 V	5.5 V	GND	"	"	B_0		"	"
			50	"							"	5.5 V	"	GND	5.5 V	"	"	"	"	A_0		"	"
			51	"							"	GND	"	5.5 V	GND	"	"	"	"	B_1		"	"
			52	"							"	"	"	5.5 V	5.5 V	"	"	"	"	A_1		"	"
			53	"							"	"	"	GND	"	"	5.5 V	"	"	A_2		"	"
			54	"							"	"	"	GND	"	"	5.5 V	"	"	B_2		"	"
			55	5.5 V							"							"	"	A_3		"	"
	I_{IL1}	3009	56	5.5 V	0.5 V						"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	A < B IN	-1.0	-2.0	mA
			57	5.5 V			0.5 V				"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	A > B IN	-1.0	-2.0	"
	I_{IL2}		58	0.5 V							"							"	"	B_3	-3.0	-6.0	"
			59	5.5 V		0.5 V					"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	A = B IN	"	"	"
			60	"							"	0.5 V	5.5 V	"	"	"	"	"	"	B_0	"	"	"
			61	"							"	5.5 V	0.5 V	"	"	"	"	"	"	A_0	"	"	"
			62	"							"	"	5.5 V	0.5 V	"	"	"	"	"	B_1	"	"	"
			63	"							"	"	"	5.5 V	0.5 V	"	"	"	"	A_1	"	"	"
			64	"							"	"	"	"	5.5 V	0.5 V	"	"	"	A_2	"	"	"
			65	"							"	"	"	"	"	5.5 V	0.5 V	"	"	B_2	"	"	"
			66	"							"	"	"	"	"	"	5.5 V	0.5 V	"	A_3	"	"	"
	I_{OS}	3011	67	GND	GND	GND		GND			"	GND	GND	GND	GND	GND	GND	GND	"	A > B OUT	-40	-100	"
			68	"		2.0 V			GND		"	"	"	"	"	"	"	"	"	A = B OUT	"	"	"
			69	"		GND	GND			GND	"	"	"	"	"	"	"	"	"	A < B OUT	"	"	"
	I_{CC}	3005	70	4.5 V	4.5 V	GND	4.5 V				"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	V_{CC}		115	"
	I_{CEX}		71	GND	GND	GND		5.5 V			"	GND	GND	GND	GND	GND	GND	GND	"	A > B OUT		250	µA
			72	"		5.5 V			5.5 V		"	"	"	"	"	"	"	"	"	A = B OUT		"	"
			73	"		GND	GND			5.5 V	"	"	"	"	"	"	"	"	"	A < B OUT		"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 1/ X and Z / Test no.	1 / 2 / B_3	2 / 3 / A < B IN	3 / 4 / A = B IN	4 / 5 / A > B IN	5 / 7 / A > B OUT	6 / 8 / A = B OUT	7 / 9 / A < B OUT	8 / 10 / GND	9 / 12 / B_0	10 / 13 / A_0	11 / 14 / B_1	12 / 15 / A_1	13 / 17 / A_2	14 / 18 / B_2	15 / 19 / A_3	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted. V_{IL} = 0.7 V and V_{OL} (max) = 0.45 V																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional test	3014 2/	74	0.8 V	2.0 V	0.8 V	0.8 V	H 3/	L 3/	L 3/	GND	2.0 V	0.8 V	2.0 V	0.8 V	0.8 V	2.0 V	2.0 V	4.5 V	2/			
			75	2.0 V	0.8 V	"	2.0 V	L	"	H	"	0.8 V	2.0 V	0.8 V	2.0 V	2.0 V	0.8 V	0.8 V	"				
			76	"	2.0 V	"	0.8 V	H	"	L	"	2.0 V	0.8 V	2.0 V	0.8 V	2.0 V	0.8 V	2.0 V	"				
			77	"	0.8 V	"	2.0 V	L	"	H	"	0.8 V	2.0 V	0.8 V	2.0 V	0.8 V	2.0 V	"	"				
			78	"	2.0 V	"	0.8 V	H	"	L	"	2.0 V	0.8 V	0.8 V	2.0 V	2.0 V	"	"	"				
			79	"	0.8 V	"	2.0 V	L	"	H	"	0.8 V	2.0 V	2.0 V	0.8 V	"	"	"	"				
			80	"	2.0 V	"	0.8 V	H	"	L	"	0.8 V	2.0 V	2.0 V	2.0 V	"	"	"	"				
			81	"	0.8 V	"	2.0 V	L	"	H	"	2.0 V	0.8 V	"	"	"	"	"	"				
			82	"	0.8 V	"	2.0 V	H	"	L	"	"	2.0 V	"	"	"	"	"	"				
			83	"	"	2.0 V	0.8 V	L	H	"	"	"	"	"	"	"	"	"	"				
			84	"	"	2.0 V	2.0 V	H	H	"	"	"	"	"	"	"	"	"	"				
			85	"	2.0 V	0.8 V	0.8 V	"	L	H	"	"	"	"	"	"	"	"	"				
			86	"	"	0.8 V	2.0 V	"	L	L	"	"	"	"	"	"	"	"	"				
			87	"	"	2.0 V	0.8 V	"	H	"	"	"	"	"	"	"	"	"	"				
			88	"	"	2.0 V	2.0 V	"	H	"	"	"	"	"	"	"	"	"	"				
			89	"	0.8 V	0.8 V	0.8 V	H	L	H	"	"	"	"	"	"	"	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 (Fig. 5)	90	IN		GND	GND			OUT	GND	GND	GND	GND	GND	GND	GND	2.7 V	5.0 V	B_3 to A < B	2	19	ns
			91	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B_0 to A < B	"	"	"
			92	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A_0 to A > B	"	"	"
			93	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B_1 to A < B	"	"	"
			94	"	GND	"		OUT			"	"	"	2.7 V	IN	GND	"	"	"	A_1 to A > B	"	"	"
			95	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A_2 to A > B	"	"	"
			96	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B_2 to A < B	"	"	"
			97	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A_3 to A > B	"	"	"
	t_{PLH2}		98	IN		2.7 V			OUT		"	"	"	"	"	"	"	GND	"	B_3 to A = B	"	21	"
			99	GND		"			"		"	IN	"	"	"	"	"	"	"	B_0 to A = B	"	"	"
			100	"		"			"		"	GND	IN	"	"	"	"	"	"	A_0 to A = B	"	"	"
			101	"		"			"		"	"	GND	IN	"	"	"	"	"	B_1 to A = B	"	"	"
			102	"		"			"		"	"	"	GND	IN	"	"	"	"	A_1 to A = B	"	"	"
			103	"		"			"		"	"	"	"	GND	IN	"	"	"	A_2 to A = B	"	"	"
			104	"		"			"		"	"	"	"	"	GND	IN	"	"	B_2 to A = B	"	"	"
			105	"		"			"		"	"	"	"	"	GND	GND	IN	"	A_3 to A = B	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 17 X and Z / Test no.	1 / 2 / B_3	2 / 3 / A < B IN	3 / 4 / A = B IN	4 / 5 / A > B IN	5 / 7 / A > B OUT	6 / 8 / A = B OUT	7 / 9 / A < B OUT	8 / 10 / GND	9 / 12 / B_0	10 / 13 / A_0	11 / 14 / B_1	12 / 15 / A_1	13 / 17 / A_2	14 / 18 / B_2	15 / 19 / A_3	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PLH3}	3003 (Fig. 5)	106	GND	IN	GND		OUT			GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	A < B to A > B	2	9.5	ns
			107	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A = B to A > B	"	"	"
	t_{PLH5}		108	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A = B to A < B	"	"	"
			109	"		GND	IN			OUT	"	"	"	"	"	"	"	"	"	A < B to A > B	"	"	"
	t_{PLH4}		110	"		IN			OUT		"	"	"	"	"	"	"	"	"	A = B to A = B	"	13	"
	t_{PHL1}		111	IN		GND	GND			OUT	"	"	"	"	"	"	"	2.7 V	"	B_3 to A < B	"	19.5	"
			112	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B_0 to A < B	"	"	"
			113	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A_0 to A > B	"	"	"
			114	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B_1 to A < B	"	"	"
			115	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A_1 to A > B	"	"	"
			116	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A_2 to A > B	"	"	"
			117	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B_2 to A < B	"	"	"
			118	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A_3 to A > B	"	"	"
	t_{PHL2}		119	14		2.7 V			OUT		"	"	"	"	"	"	"	GND	"	B_3 to A = B	"	"	"
			120	GND		"			"		"	IN	"	"	"	"	"	"	"	B_0 to A = B	"	"	"
			121	"		"			"		"	GND	IN	"	"	"	"	"	"	A_0 to A = B	"	"	"
			122	"		"			"		"	"	GND	IN	"	"	"	"	"	B_1 to A = B	"	"	"
			123	"		"			"		"	"	"	GND	IN	"	"	"	"	A_1 to A = B	"	"	"
			124	"		"			"		"	"	"	"	GND	IN	"	"	"	A_2 to A = B	"	"	"
			125	"		"			"		"	"	"	"	"	GND	IN	"	"	B_2 to A = B	"	"	"
			126	"		"			"		"	"	"	"	"	"	GND	IN	"	A_3 to A = B	"	"	"
	t_{PHL3}		127	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A < B to A > B	"	10.5	"
			128	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A = B to A > B	"	"	"
	t_{PHL5}		129	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A = B to A < B	"	"	"
			130	"		GND	IN			OUT	"	"	"	"	"	"	"	"	"	A < B to A > B	"	"	"
	t_{PHL4}		131	"		IN			OUT		"	"	"	"	"	"	"	"	"	A = B to A > B	"	9.5	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases J and X and Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	B_3	A < B IN	A = B IN	A > B IN	A > B OUT	A = B OUT	A < B OUT	GND	B_0	A_0	B_1	A_1	A_2	B_2	A_3	V_{CC}				
10 T_C = +125°C	t_{PLH1}	3003 (Fig. 5)	132	IN		GND	GND			OUT	GND	GND	GND	GND	GND	GND	GND	2.7 V	5.0 V	B_3 to A < B	2	23	ns
			133	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B_0 to A < B	"	"	"
			134	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A_0 to A > B	"	"	"
			135	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B_1 to A < B	"	"	"
			136	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A_1 to A > B	"	"	"
			137	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A_2 to A > B	"	"	"
			138	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B_2 to A < B	"	"	"
			139	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A_3 to A > B	"	"	"
	t_{PLH2}		140	IN		2.7 V			OUT		"	"	"	"	"	"	"	GND	"	B_3 to A = B	"	25	"
			141	GND		"			"		"	IN	"	"	"	"	"	"	"	B_0 to A = B	"	"	"
			142	"		"			"		"	GND	IN	"	"	"	"	"	"	A_0 to A = B	"	"	"
			143	"		"			"		"	"	GND	IN	"	"	"	"	"	B_1 to A = B	"	"	"
			144	"		"			"		"	"	"	GND	IN	"	"	"	"	A_1 to A = B	"	"	"
			145	"		"					"	"	"	"	GND	IN	"	"	"	A_2 to A = B	"	"	"
			146	"		"					"	"	"	"	"	GND	IN	"	"	B_2 to A = B	"	"	"
			147	"		"					"	"	"	"	"	"	GND	IN	"	A_3 to A = B	"	"	"
	t_{PLH3}		148	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A < B to A > B	"	11.5	"
			149	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A = B to A > B	"	"	"
	t_{PLH5}		150	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A = B to A < B	"	"	"
			151	"		GND	IN			OUT	"	"	"	"	"	"	"	"	"	A > B to A < B	"	"	"
	t_{PLH4}		152	"		IN			OUT		"	"	"	"	"	"	"	"	"	A = B to A = B	"	16	"
	t_{PHL1}		153	IN		GND	GND			OUT	"	"	"	"	"	"	"	2.7 V	"	B_3 to A < B	"	24	"
			154	GND		"	GND			OUT	"	IN	2.7 V	"	"	"	"	GND	"	B_0 to A < B	"	"	"
			155	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A_0 to A > B	"	"	"
			156	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B_1 to A < B	"	"	"
			157	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A_1 to A > B	"	"	"
			158	"	GND	"		OUT			"	"	"	GND	GND	IN	2.7 V	"	"	A_2 to A > B	"	"	"
			159	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B_2 to A < B	"	"	"
			160	2.7 V	GND	"		OUT			"	"	"	"	"	"	GND	GND	IN	"	A_3 to A > B	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 1/ X and 2 / Test no.	1 / 2 / B_3	2 / 3 / A < B IN	3 / 4 / A = B IN	4 / 5 / A > B IN	5 / 7 / A > B OUT	6 / 8 / A = B OUT	7 / 9 / A < B OUT	8 / 10 / GND	9 / 12 / a_0	10 / 13 / A_0	11 / 14 / B_1	12 / 15 / A_1	13 / 17 / A_2	14 / 18 / B_2	15 / 19 / A_3	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	t_{PHL2}	3003 (Fig. 5)	161	IN		2.7 V			OUT		GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	B_3 to A = B	2	24	ns
			162	GND		"			"		"	IN	"	"	"	"	"	"	"	B_0 to A = B	"	"	"
			163	"		"			"		"	GND	IN	"	"	"	"	"	"	A_0 to A = B	"	"	"
			164	"		"			"		"	"	GND	IN	"	"	"	"	"	B_1 to A = B	"	"	"
			165	"		"			"		"	"	"	GND	IN	"	"	"	"	A_1 to A = B	"	"	"
			166	"		"			"		"	"	"	"	GND	IN	"	"	"	A_2 to A = B	"	"	"
			167	"		"			"		"	"	"	"	"	GND	IN	"	"	B_2 to A = B	"	"	"
			168	"		"			"		"	"	"	"	"	"	GND	IN	"	A_3 to A = B	"	"	"
	t_{PHL3}		169	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A < B to A > B	"	13	"
			170	"	GND	IN		OUT			"	"	"	"	"	"	"	"	"	A = B to A > B	"	"	"
	t_{PHL5}		171	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A = B to A < B	"	"	"
			172	"		GND	IN			OUT	"	"	"	"	"	"	"	"	"	A > B to A < B	"	"	"
	t_{PHL4}		173	"		IN			OUT		"	"	"	"	"	"	"	"	"	A = B to A = B	"	12	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C																						

1/ Cases X and 2 pins not designated are NC.

2/ Tests shall be performed in sequence.

3/ Output voltages shall be either (a) H = 2.5 V, minimum and L = 0.5 V, maximum when using a high speed checker double comparator; or (b) H ≥ 1.5 V when using a high speed checker single comparator.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IN}	Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54S85

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6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit	
	A	B
	Texas Instruments	Signetics Corporation
01	X	X

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1078)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/82B	MICROCIRCUITS, DIGITAL, SCHOTTKY TTL, MAGNITUDE COMPARATORS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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